Dictionary learning using sparse autoencoders. Exhaustive search for meaningful non-basis-aligned directions is impossible due to the infinite search space. The *dictionary learning* literature tackles this problem by performing an unsupervised search for directions in neuron activations which both (1) capture the information encoded in the internal representations and (2) are *disentangled* from other meaningful directions. Bengio, Courville, and Vincent (2013) characterize disentangled representations as *factors of variation* in the training dataset.

To identify these factors of variation, Sharkey, Braun, and Millidge (2023) used sparse autoencoders (SAEs) to perform dictionary learning on a one-layer transformer, identifying a large (overcomplete) basis of features. Given activation vector $\mathbf{h}^\ell$, SAEs are trained to reconstruct $\mathbf{h}^\ell$ as $\hat{\mathbf{h}}^\ell$ while only activating a sparse subset of dictionary features. Concretely, SAEs typically consist of an encoder and decoder:

$$\mathbf{f} = \mathrm{ReLU}(W_e(\mathbf{h}^\ell - \mathbf{b}_d) + \mathbf{b}_e) \tag{3}$$

$$\hat{\mathbf{h}}^\ell = W_d\mathbf{f} + \mathbf{b}_d, \tag{4}$$

where $\mathbf{f}$ is the feature vector (the encoded space), W denotes a learned weight matrix, and $\mathbf{b}$ denotes a learned bias vector.[27] It is common to refer to a single dimension of $\mathbf{f}$, or $\mathbf{f}_i$, as a *feature*. Cunningham et al. (2024) applied SAEs to language models and demonstrated that the observed dictionary features are highly interpretable and can be used to localize and edit model behavior. Since then, numerous researchers have found promising results in identifying funtionally relevant and human-interpretable features (Templeton et al. 2024; Rajamanoharan et al. 2024; Braun et al. 2024; Bricken et al. 2023; Fel et al. 2025, *inter alia*), many of them having predictable effects on the model's behavior under interventions. That said, SAEs are not able to perfectly reconstruct the model's activations, and may not be optimal for counterfactual operations such as steering (Wu et al. 2025). Most importantly, however, we do not know *a priori* what the ground truth features are in the model's computation, and can only use the reconstruction performance as a proxy measure of performance.

Correlation-based clustering. Another unsupervised way of discovering meaningful units is clustering mediators by the similarity of their behavior over some dataset $\mathcal{D}$. This idea is not new (cf. Elman 1990), but running causal verifications of the qualitative insights from clustering studies is relatively rare. Bau et al. (2019a) showed that neurons sharing a similar behavior are causally important for various functionalities in recurrent neural machine translation models. Dalvi et al. (2020) cluster neurons in language models, and are able to maintain performance after ablating a significant portion of them. The goal of Dalvi et al.'s study was not interpretability, but their results nonetheless causally verify that redundancy is very common in neural networks.

There has recently been renewed interest in mediator search via clustering. Michaud et al. (2023) propose a method to identify interpretable behaviors within neural networks by clustering parameters. Because the identified behaviors tend to be coherent, the units implicated in each cluster can be viewed as a set of components that have a functionally coherent role in the network. Marks et al. (2025) and Engels et al. (2025) generalize this from gradients to neuron or sparse autoencoder activations. The activations

27 This is a basic formulation of SAEs. There are more sophisticated architectures that have empirically demonstrated better reconstruction performance and/or more interpretable features (e.g., Rajamanoharan et al. 2024; Braun et al. 2024).

that compose the clusters are then labeled according to the dataset samples on which they activate most highly. When based on neurons, clusters are basis-aligned subspaces; when based on sparse autoencoder features (as in Marks et al. 2025), they are non-basis-aligned subspaces. To compute the indirect effect of a cluster, one can intervene on each component in the cluster simultaneously and compute the resulting indirect effect. Then, to find the most causally relevant clusters, one can exhaustively search by taking the top cluster by indirect effect. This method has not yet been extensively employed or explored. However, intervening on elements within these clusters could be a useful way to establish the functional role of *groups of* components in future work, or assess whether a subset of a model's behavior is implicated in a more complex task. We discuss this in §7.

In summary, unsupervised mediator searches enable us to locate human-interpretable concepts from neural networks internals without any labeled data. They allow us to do so relatively quickly, and these concepts can then be causally implicated in model behavior post hoc. However, the recovered concepts are not guaranteed to be faithful to the model's true concept space, and we are never guaranteed to recover a complete nor non-redundant concept set. Another pressing issue in unsupervised search is evaluation: while it is possible to compare the relative quality of two different unsupervised interpretability searches, it is difficult to devise absolute metrics that meaningfully capture closeness to some ideal solution; indeed, an ideal or identifiable solution may not exist (Méloux et al. 2025). For SAEs, some have tried to devise thorough evaluations (Karvonen et al. 2025), but robust evaluations that enable comparisons between supervised and unsupervised methods have only just started to become common (e.g., Wu et al. 2025; Mueller et al. 2025).

7. Discussion

What is the right unit of analysis for describing the inner workings of neural networks? Thus far, we have categorized past work by mediator type and search method. Now, we turn our attention to practical questions: What kinds of studies can more easily be done with certain mediator types? Where is there still room to deploy less common mediators in useful ways? Our main goal is to encourage researchers to conceptualize the field in this way, in the hope that we can encourage more work on discovering better causal abstractions and better terms for discussing the inner workings of neural machine learning systems—and, more ambitiously, more rigorous theoretical foundations for interpretability.

7.1 What is the right mediator?

There are pros and cons to any mediator, and the best mediator will therefore depend on one's goals. In this section, we ask: When is it appropriate to deploy particular kinds of mediators and search methods? To answer this question, we revisit the three goals laid out in §2.1 and the pros and cons discussed in §5 and §6. We first hypothesize which mediators and search methods are likely to be best given recent evidence. Then, in the following subsections, we describe directions for future work that can further improve on these criteria, and call for work formalizing these criteria into concrete benchmarks. In Appendix D, we provide an even more practical guide by giving concrete examples in specific research scenarios.

Explaining model behavior. If we wish to understand at an algorithmic level how a model performs some behavior, then in the absence of compute restrictions and with no strong prior hypotheses, **unsupervised optimization-based methods** (§6.2.2) over fine-grained mediators (such as **non-basis-aligned directions**, §5.3) provide a strong starting point. For example, unsupervised methods like sparse autoencoders provide a fine-grained (selective) and human-interpretable interface to a model's computation, making explanations easier to derive in the absence of any pre-existing mechanistic hypotheses. They have also been found to yield higher faithfulness with fewer components compared to components like neurons (Marks et al. 2025), meaning that non-basis-aligned directions generally achieve a better trade-off between sparsity and faithfulness than basis-aligned directions. That said, autoencoder features are not guaranteed to be faithful to a neural network's behavior in next-token prediction, and they require either a human or an LLM to label or interpret the features, which is laborious and expensive (Bills et al. 2023; Paulo et al. 2024). Moreover, natural language explanations of model components have inherent flaws (Huang et al. 2023): they may often exhibit both low precision and recall. Finally, non-basis-aligned directions, while more interpretable than basis-aligned components, require more human effort and/or compute than basis-aligned components to locate, and one may need to rediscover these directions if model fine-tuning, adaptation, or editing is part of the study.[28]

Verifying a mechanistic hypothesis. If we already have a hypothesis as to how a model performs some behavior and wish to measure the accuracy of our hypothesis, then **multidimensional non-basis-aligned subspaces** (§5.3) may be the right mediators, and a reasonable corresponding search method would be **counterfactual-based optimization** (§6.2.1). One can automatically search for the subspaces which correspond to a particular node in one's hypothesized causal graph using alignment search methods (Geiger et al. 2024; Wu et al. 2023; Sun et al. 2025; Geiger et al. 2021). Alignment search entails learning a linear transformation to isolate some target concept; this allows us to locate distributed representations that act as independent causal variables in non-basis-aligned spaces. This is relatively scalable, and enables us to qualitatively understand intermediate model computations. The primary downside is that we must anticipate the mechanisms that models employ to perform a task, as demonstrated in Appendix C; if we cannot anticipate them, then curating data and refining one's causal hypotheses may require significant human effort. If these are significant concerns, then probing may be a better (though correlative and less precise) option. These methods are subject to the same confounds as any other optimization-based technique: the module we are optimizing could directly learn the phenomenon, rather than extracting it from the model (Hewitt and Liang 2019; Sutter et al. 2025).

Localization and editing. If one's goal is to localize some phenomenon in a model, then **exhaustive searches** (§6.1) over **basis-aligned subspaces** (§5.2) or **full layers** (§5.1) may be sufficient. There are many comprehensive causal techniques for locating these, including causal tracing (Meng et al. 2022) and activation patching (Vig et al. 2020), as well as techniques for locating graphs of basis-aligned mediators, such as circuit discovery algorithms (Goldowsky-Dill et al. 2023; Wang et al. 2023; Conmy et al. 2023). Some of these methods are slow in their exact form, but fast approximations exist to

28 Though Prakash et al. (2024) find that the same model components are implicated in an entity tracking task before and after fine-tuning.